

ASSIGNMENT

Automated soldiers :

1) PERFORMANCE

- Mission success
- Survival Rate
- Threat Neutralisation
- Minimal Collateral Damage

2) ENVIRONMENT

- Battlefield
- Urban Zones
- Terrain with enemies & allies

3) ACTUATORS

- Weapons
- Movement (Arms/Legs)
- Communication Modules

4) SENSORS

- Cameras
- IR sensors
- RADAR
- GPS
- Audio sensors

Playing a Tennis Match :

1) PERFORMANCE

- Points won
- Match victory
- Accuracy
- Stamina efficiency

2) ENVIRONMENT

- Tennis court
- Suitable weather conditions (if outdoor)

3) ACTUATORS

- Arms (Racket swings)
- Legs
- Grip control

4) SENSORS

- Vision
- Proprioception
- Touch
- Timing sensors

Knitting a Sweater

1) PERFORMANCE

- correct Pattern
- minimal errors
- material efficiency

2) ENVIRONMENT

- Yarn
- Needles
- Pattern instructions
- Workspace

3) ACTUATORS

- Fingers
- Hands
- Wrist Movements
- Automated Needle control

4) SENSORS

- vision (Stitch tracking)
- Touch (Tension)
- Counting mechanism

Playing Soccer

1) PERFORMANCE

- Goals scored, assists, wins, fouls minimised

2) ENVIRONMENT

- Soccer field, teammates, opponents, Ball, weather

3) ACTUATORS

- Legs, feet, body movement, communication

4) SENSORS

- Vision, hearing, balance, touch

Exploring the subsurface oceans of Titan

1) PERFORMANCE

- Data collected, mission duration, system safety

2) ENVIRONMENT

- Subsurface ocean, extreme pressure, darkness, unknown terrain

3) ACTUATORS

- Propellers, Robotic arms, sampling Tools

4) SENSORS

- Sonar, chemical sensors, cameras, Pressure sensors

Shopping for used AI Books on the Internet

1) PERFORMANCE

- Books Relevance, Price Minimisation, Purchase Success

2) ENVIRONMENT

- Websites, sellers, Auctions, search engines

3) ACTUATORS

- Mouse clicks, keyboard i/p, Payment Actions

4) SENSORS

- Screen display, Price listings, Reviews, Availability info.

Practicing Tennis Against a Wall

1) PERFORMANCE

- Rally length, shot accuracy, consistency

2) ENVIRONMENT

- Wall, court, Ball

3) ACTUATORS

- Arms, legs, wrist Movement

4) SENSORS

- Vision, Timing, Balance, Muscle Feedback

Bidding on an Item at an Auction

1) PERFORMANCE

- Item won, lowest possible price, timing accuracy

2) ENVIRONMENT

- Auction Platform, competing bidders, Time constraints

3) ACTUATORS

- Mouse clicks, Keyboard Input, bid submission

4) SENSORS

- Current bid price, countdown timer, competitor bids

Performing a High Jump

1) PERFORMANCE

- Height cleared, Technique efficiency, safety

2) ENVIRONMENT

- Runway, bar, landing mat

3) ACTUATORS

- Legs, arms, core muscles

4) SENSORS

- Vision, Balance, Muscle feedback